

Foundation (Jalapeno)

Q1. Solve for x in the equation $2^x = 8$

- (A) $x = 1$
- (B) $x = 2$
- (C) $x = 3$
- (D) $x = 4$
- (E) $x = 5$

Q2. If $3^x = 27$, what is the value of x ?

- (A) $x = 1$
- (B) $x = 2$
- (C) $x = 3$
- (D) $x = 4$
- (E) $x = 5$

Q3. A basketball player scores 2^x points in a game. If she scores 16 points, what is the value of x ?

- (A) $x = 2$
- (B) $x = 3$
- (C) $x = 4$
- (D) $x = 5$
- (E) $x = 6$

Q4. Solve for y in the equation $4^y = 64$

- (A) $y = 1$
- (B) $y = 2$
- (C) $y = 3$
- (D) $y = 4$
- (E) $y = 5$

Q5. A cyclist travels 3^x miles in a day. If she travels 9 miles, what is the value of x ?

- (A) $x = 1$
- (B) $x = 2$
- (C) $x = 3$
- (D) $x = 4$
- (E) $x = 5$

Q6. If $2^x = 32$, what is the value of x ?

- (A) $x = 2$
- (B) $x = 3$
- (C) $x = 4$
- (D) $x = 5$
- (E) $x = 6$

Q7. Solve for z in the equation $5^z = 125$

- (A) $z = 1$
- (B) $z = 2$
- (C) $z = 3$
- (D) $z = 4$
- (E) $z = 5$

Q8. A football team wins 2^x games in a season. If they win 8 games, what is the value of x ?

- (A) $x = 1$
- (B) $x = 2$
- (C) $x = 3$
- (D) $x = 4$
- (E) $x = 5$

Open Ended Questions (FRQ)

Q9. Solve for x in the equation $x^2 = 2^x$ and show your work

Q10. If a runner's distance is given by $d = 3^x$, and she runs 27 miles, find the value of x and show your calculation

Chef's Tips

- Read each question carefully
- Choose the correct answer for multiple-choice questions
- Show your work for free-response questions